# Quantifying precessional effects on binary black-hole systems by introducing a generic spinning waveform template

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## I. INTRODUCTION

One of the main predictions of general relativity is the emission of GWs from compact binary coalescence (CBC) processes. After almost a century of attempts, the first successful observation of GWs from a binary black hole (BBH) merger was reported from the LIGO and the VIRGO collaboration in 2016 [1]. Advanced LIGO (aLIGO) [2] and advanced VIRGO (aVIRGO) [3] performed two more runs after the first successful one, and brought the total number of GW detections to 90 events [4]. Out of these 90 events, 84 are confirmed to be from a BBH merger, 4 of them from neutron star black hole mergers and the last 2 were potentially produced from a binary neutron star merger. Recently, Kamioka Gravitational Wave Detector (KAGRA) [5–7] has become operational and will join the aLIGO and aVIRGO to form the LVK detection network for future observational runs. Hundreds (millions) of additional events are expected in the next years (decades) due to the continuous improvements in the sensitivity of the LVK network, together with the proposed next generation of detectors, such as the Einstein Telescope [8], the Cosmic Explorer (CE) [9], the Deci-Hertz Interferometer Gravitational Wave Observatory (DECIGO) [10] and the Laser Interferometer Space Antenna (LISA) [11].

In the most recent O3b run, the LVK collaboration used four pipelines to search for GW signals. Out of the four, three of them, GstLAL [12–14], Multi-band template analysis [15, 16] and PyCBC [17, 18] perform their searches by using already constructed CBC waveform templates, while the remaining method, cWB [19, 20] is doing so by making minimal assumption about the behavior of the sources. The CBC search pipelines utilize match-filter analysis [21, 22] to compare the signal received on the ground-based detectors with a bank of carefully constructed templates as a means to extract the information embedded in the signal about the source's properties. These templates strongly depend on the individual masses of the BHs as well as their spin components in order to accurately model the GW signals. Even though the masses of the BHs and some aligned combinations of their spins are well constrained, the short duration of the signals as well as the subtle effects of the spins on the templates make them difficult to measure individually (LIGO CITE).

Current observations of BBH mergers confirm that half of the individual spin magnitudes lie below $\chi_i \approx 0.26$ are typically misaligned with the orbital angular momentum $\mathbf{L}$ [23]. In turn, $\mathbf{L}$ also precesses and nutates around the total angular momentum $\mathbf{J}$, whose direction remains approximately constant during the inspiral part of the CBC [24, 25]. This spin and orbital precession plays a key role in BBH dynamics [24] by imprinting modulations to the amplitude and the phase of the GW signal [26].

These modulations enter the GW templates at the 1.5PN and 2PN order as spin-orbit and spin-spin coupling effects respectively [24, 26–28]. State-of-the-art waveform model families used in GW data analysis, such as IMRPhenomXPHM [29], SEOBNRv4PHM [30], and NRSur7dq4 [31], model these effects accurately. However, it is fairly challenging to analytically quantify how the two spin vectors generically affect the GW template. As current detectors cannot unambiguously constrain both spins, see e.g., [32], the two spins are typically represented by combinations of the BBH masses and spins that approximate the effects. Two main such parameters are the effective inspiral spin $\chi_{\rm eff}$ [33] which is conserved through the PN inspiral [34] of the coalescence and influences its length, and the effective precession spin $\chi_{\rm p}$ [35–37] which approximates the in-plane spin component responsible for

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spin precession.

In our previous analysis [38], we developed a different set of phenomenological parameters that encompass the essential features of spin precession and nutation. These five parameters provide a geometric framework to characterize the various configurations in which the evolution of $\mathbf{L}$ about $\mathbf{J}$ impact the BBH dynamics. In our current analysis, we take one step forward and introduce the regular precessing parameters, the precession amplitude $\theta_{LJ}$ and precession frequency $\Omega_{LJ}$, as functions of the GW frequency into a new 'toy' model GW template. We then show how these parameters imprint a distinct signature on the template's various amplitude and phase terms and finally we try to extract the signal-to-noise ratios needed to detect precession in the next LVK run as well as in future 3G detectors. MK: Can we answer this last question with the single-detector response we consider?

NAS: Maybe need to emphasize upfront that we're presenting an inspiral-only waveform model and not trying to "compete" with full inspiral-merger-ringdown models.

This paper is organized as follows.ES:fill this up

## II. BASIC FORMALISMS AND NOTATIONS

In this section, we briefly review the inspiral waveform in the Newtonian, quadrupole-moment approximation [24, 39, 40] and our taxonomy of spin precession [38]. We will incorporate the latter into the former in the following section to construct a toy model of a precessing frequency-domain gravitational waveform.

### A. Gravitational waveform in the quadrupole-moment approximation

In the frequency domain, a GW signal observed by a ground-based detector can be expressed as [24, 26, 40]

$$\tilde{h}(f) = F_+ \tilde{h}_+(f) + F_\times \tilde{h}_\times(f)\,, \tag{1}$$

where $F_+$ and $F_\times$ are the detector beam-pattern coefficients and $h_+$ and $h_\times$ are the projections of the GW field onto the two principal polarization states. For an L-shaped GW detector, these beam-pattern coefficients can be expressed as [24, 41]

$$\begin{aligned} F_+(\theta_S,\Phi_S,\psi) &= \frac{1}{2}(1+\cos^2\theta_S^2)\cos 2\Phi_S \cos 2\psi \\ &\quad - \cos\theta_S \sin 2\Phi_S \sin 2\psi\,, \\ &= C\cos(2\psi+\alpha)\,, \end{aligned} \tag{2a}$$

$$\begin{aligned} F_\times(\theta_S,\Phi_S,\psi) &= \frac{1}{2}(1+\cos^2\theta_S)\cos 2\Phi_S \sin 2\psi \\ &\quad + \cos\theta_S \sin 2\Phi_S \cos 2\psi\,, \\ &= C\sin(2\psi+\alpha)\,, \end{aligned} \tag{2b}$$

where the spherical angles $\theta_S, \Phi_S$ specify the sky location $\hat{\mathbf{N}}$ of the source with respect to the detector frame $\{\hat{\mathbf{X}}_D, \hat{\mathbf{Y}}_D, \hat{\mathbf{Z}}_D\}$ defined with respect to the arms of the L-shaped detector. This frame, along with the sky and source frames we will use later, are described in greater detail in Appendix A.

The polarization angle $\psi$ between the principal + direction and the direction of constant azimuth is given by [24]

$$\psi = \tan^{-1}\left[\frac{\hat{\mathbf{L}}\cdot\hat{\mathbf{Z}}_D - (\hat{\mathbf{L}}\cdot\hat{\mathbf{N}})(\hat{\mathbf{Z}}_D\cdot\hat{\mathbf{N}})}{\hat{\mathbf{N}}\cdot(\hat{\mathbf{L}}\times\hat{\mathbf{Z}}_D)}\right], \tag{3}$$

where $\hat{\mathbf{L}}$ is the unit vector in the direction of the BBH orbital angular momentum. The beam-pattern amplitude and phase are

$$C = \left[\frac{1}{4}(1+\cos\theta_S^2)^2\cos^2 2\Phi_S + \cos^2\theta_S \sin^2 2\Phi_S\right]^{1/2}, \tag{4a}$$

$$\alpha = \tan^{-1}\left(\frac{2\cos\theta_S \tan 2\Phi_S}{1+\cos\theta_S^2}\right). \tag{4b}$$

The beam-pattern amplitude has a range $0 \leq C \leq 1$ and saturates the upper (lower) inequality for $\hat{\mathbf{N}} = \pm\hat{\mathbf{Z}}_D$ [$\hat{\mathbf{N}} = 2^{-1/2}(\pm\hat{\mathbf{X}}_D \pm \hat{\mathbf{Y}}_D)$].

The frequency-domain GW polarizations are [26, 40]

$$\tilde{h}_+(f) = Af^{-7/6}[1+(\hat{\mathbf{L}}\cdot\hat{\mathbf{N}})^2]e^{i\Psi}\,, \tag{5a}$$

$$\tilde{h}_\times(f) = -2iAf^{-7/6}(\hat{\mathbf{L}}\cdot\hat{\mathbf{N}})e^{i\Psi}\,, \tag{5b}$$

where $A$ is

$$A = \sqrt{\frac{5}{96}}\frac{1}{D_L}\frac{\mathcal{M}^{5/6}}{\pi^{2/3}}\,, \tag{6}$$

$D_L$ is the luminosity distance between the detector and the source and $\mathcal{M}$ is the chirp mass. For a binary with total mass $M = m_1 + m_2$ and mass ratio $q = m_2/m_1 \leq 1$, the symmetric mass ratio is $\eta = q/(1+q)^2$ and the chirp mass is $\mathcal{M} = \eta^{3/5}M$. The spin-independent GW phase at 2PN order is [42]

$$\begin{aligned} \Psi(f) &= 2\pi f t_c - \phi_c - \frac{\pi}{4} + \frac{3}{128}\eta^{-1}u^{-5} \\ &\quad \times\left[1+\frac{20}{9}\left(\frac{743}{336}+\frac{11}{4}\eta\right)u^2 - 16\pi u^3 + 10\Gamma u^4\right], \end{aligned} \tag{7}$$

where $u \equiv (\pi M f)^{1/3} = (M/r)^{1/2}$ is our PN parameter, $t_c$ and $\phi_c$ are the time and GW phase at binary coalescence, and

$$\Gamma \equiv \frac{3058673}{1016064} + \frac{5429}{1008}\eta + \frac{617}{114}\eta^2\,. \tag{8}$$

Inserting Eqs. (5) into (1) yields

$$\tilde{h}(f) = Be^{i[\Psi(f)-\phi_p+2\delta\Phi]}\,, \tag{9}$$

where the GW amplitude $B$ is given by

$$\begin{aligned} B &= Af^{-7/6}\{4(\hat{\mathbf{L}}\cdot\hat{\mathbf{N}})^2F_\times^2 + [1+(\hat{\mathbf{L}}\cdot\hat{\mathbf{N}})^2]^2F_+^2\}^{1/2}\,, \\ &= ACf^{-7/6}\{4(\hat{\mathbf{L}}\cdot\hat{\mathbf{N}})^2\sin^2(2\psi+\alpha) \\ &\quad + [1+(\hat{\mathbf{L}}\cdot\hat{\mathbf{N}})^2]^2\cos^2(2\psi+\alpha)\}^{1/2}\,, \end{aligned} \tag{10}$$

the polarization phase $\phi_p$ is

$$\begin{aligned} \phi_p &= \tan^{-1}\left[\frac{2(\hat{\mathbf{L}}\cdot\hat{\mathbf{N}})F_\times}{[1+(\hat{\mathbf{L}}\cdot\hat{\mathbf{N}})^2]F_+}\right]\,, \\ &= \tan^{-1}\left[\frac{2(\hat{\mathbf{L}}\cdot\hat{\mathbf{N}})\tan(2\psi+\alpha)}{1+(\hat{\mathbf{L}}\cdot\hat{\mathbf{N}})^2}\right]\,, \end{aligned} \tag{11}$$

and $\delta\Phi$ is an additional contribution to the GW phase for precessing BBH systems given by integrating [24]

$$\frac{d\delta\Phi}{df} = \left[\frac{\hat{\mathbf{L}}\cdot\hat{\mathbf{N}}}{1-(\hat{\mathbf{L}}\cdot\hat{\mathbf{N}})^2}\right](\hat{\mathbf{L}}\times\hat{\mathbf{N}})\cdot\frac{d\hat{\mathbf{L}}}{df}\,. \tag{12}$$

As we are only interested in the inspiral portion of the waveform during which precession can occur, we cut off our waveforms at a GW frequency

$$f_{\rm cut} = \frac{1}{6^{3/2}\pi M} = 4.3\times10^3\,{\rm Hz}\left(\frac{M}{M_\odot}\right)^{-1} \tag{13}$$

equivalent to the quadrupole frequency at the innermost stable circular orbit of a BH of mass $M$ [26, 43]. NAS: To have a BBH with an inspiral in the LIGO band, it must not be too massive. Indeed, it seems the examples y'all look at are low mass. Is there a maximum mass below which this model is valid for given frequency limits? I.e., GW150921 was basically just a merger+ringdown in the LIGO band. ES:I think Fig.4 is trying to show exactly that. earlier versions of the plot used to have a solid vertical line that denoted the maximum total mass a system must have to spend at least some time in band. Mike also had a good idea about how to replot Fig.4 using shaded regions. We should talk about it next week

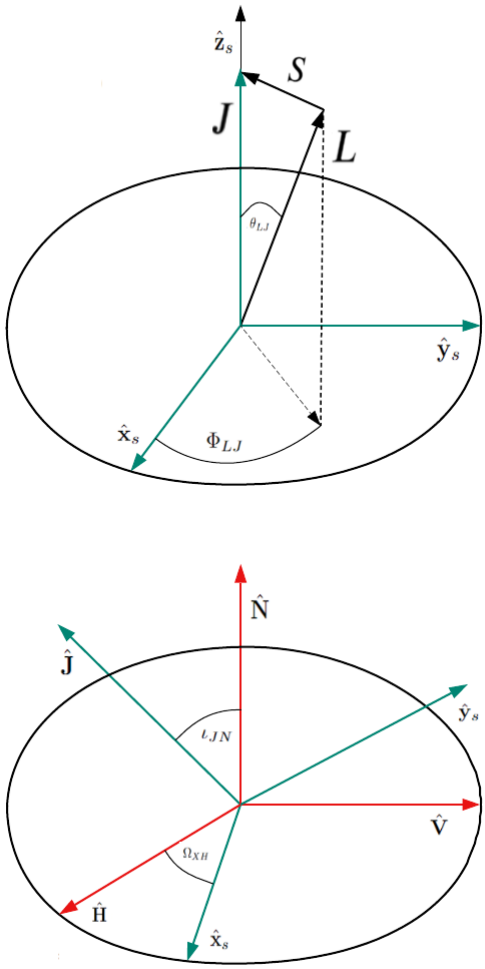



FIG. 1. Schematic depictions of the sky and source coordinate frames. The sky frame, shown in red in the bottom panel, is defined such that the unit vector $\hat{\mathbf{N}}$ pointing from the detector to the sky location of the GW source is along the $z$ axis, the unit vector $\hat{\mathbf{V}}$ points towards the zenith at the detector location is along the $y$ axis, and the unit vector $\hat{\mathbf{H}} = \hat{\mathbf{V}}\times\hat{\mathbf{N}}$ completes the orthonomal triad. The source frame is defined such that the total angular momentum $\hat{\mathbf{J}}$ is along the $Z_S$ axis, $\hat{\mathbf{N}}\times\hat{\mathbf{J}}$ is along the $X_S$ axis, and $\hat{\mathbf{Y}}_S = \hat{\mathbf{Z}}_S\times\hat{\mathbf{X}}_S$ completes the orthonormal triad.

### B. Taxonomy of BBH precession and nutation

The gravitational waveform described in the previous subsection can be determined once the frequency-dependent direction of the orbital angular momentum $\hat{\mathbf{L}}(f)$ is specified. This problem was considered in Gangardt *et al.* [38] which created a taxonomy of spin precession. Following [24], this work defined precession as *simple* when the direction of the total angular momentum $\hat{\mathbf{J}}$ is approximately constant during the inspiral. This is an excellent approximation except in the special cases of transitional precession [24] or nutational resonances [25]. In the generic case of simple precession, misaligned BBH spins make the orbital angular momentum $\mathbf{L}$ to precess about the total angular momentum $\mathbf{J}$ with polar angle $\theta_{LJ}$ and azimuthal angle $\Phi_{LJ}$ that both vary on the precession timescale. The oscillation of $\theta_{LJ}$ on the precession timescale is called *nutation*. In the special case of regular precession, nutation vanishes and both $\theta_{LJ}$ and the precession frequency $\Omega_{LJ}\equiv d\Phi_{LJ}/dt$ vary on the longer radiation-reaction timescale.

Nine general coordinates would be needed to specify $\mathbf{L}$ and the individual spins $\mathbf{S}_1$ and $\mathbf{S}_2$ of a precessing BBH system. However, for the case of simple precession, Kesden *et al.* [44] recognized that, on the precession timescale, the constancy of $\mathbf{J}$, $L$, $S_1$, $S_2$, and the projected effective spin $\chi_{\rm eff}$ [33, 34] imply that such systems only possess

two degrees of freedom. These degrees of freedom can be described by one intrinsic parameter, the total spin magnitude $S = |\mathbf{S}_1 + \mathbf{S}_2|$, and one extrinsic parameter, a global rotation of the BBH system by an angle $\Phi_{LJ}$ about $\mathbf{J}$. As indicated in the top panel of Fig. 1, oscillation of $S$ between its extrema $S_-$ and $S_+$ on the precession timescale corresponds to nutation, the oscillation of

$$\theta_{LJ}(S) = \cos^{-1}\left(\frac{J^2 + L^2 - S^2}{2JL}\right) \tag{14}$$

between its extrema $\theta_{LJ\pm} = \theta_{LJ}(S_\pm)$. The duration of this oscillation defines the nutation period

$$\tau = 2\int_{S_-}^{S_+} \frac{dS}{|dS/dt|}\,. \tag{15}$$

As $S$ describes the only intrinsic degree of freedom of the system, one can define the *precession average* of any quantity $X$ varying on the precession timescale as

$$\langle X\rangle \equiv \frac{2}{\tau}\int_{S_-}^{S_+} \frac{X(S)}{|dS/dt|} dS\,. \tag{16}$$

In addition to the polar angle $\theta_{LJ}(S)$, such quantities include the precession frequency $\Omega_{LJ}(S) = d\Phi_{LJ}/dt$. Precessional dynamics in this formalism are described in much greater detail in Gerosa *et al.* [45] which also provides expressions for $S_\pm$, $dS/dt$, and $\Omega_{LJ}(S)$.

Gangardt *et al.* [38] used this formalism to describe the evolution of $\hat{\mathbf{L}}$ in terms of five new precession parameters in the generic case of simple precession: (1) the precession amplitude $\langle\theta_{LJ}\rangle$, (2) the precession frequency $\langle\Omega_{LJ}\rangle$, (3) the nutation amplitude $\Delta\theta_{LJ} = (\theta_{LJ+} - \theta_{LJ-})/2$, (4) the nutation frequency $\omega = 2\pi/\tau$, and (5) the variation of the precession frequency $\Delta\Omega_{LJ} = [\Omega_{LJ}(S_+) - \theta_{LJ}(S_-)]/2$.

These parameters have several advantages over other approaches to characterizing generic simple precession. By describing the evolution of $\hat{\mathbf{L}}$, they have a more direct connection the the inspiral waveform (9) than the individual spins $\mathbf{S}_i$. They also vary on the longer radiation-reaction timescale, making parameter estimates less sensitive to the reference frequency at which they are specified. They also provide a richer and more accurate description of generic simple precession than any single parameter like the effective precession spin $\chi_{\rm p}$ [35–37] ever could. Even the case of regular precession considered in this paper is two dimensional, being described by the precession amplitude $\langle\theta_{LJ}\rangle$ and frequency $\langle\Omega_{LJ}\rangle$. In the next section, we incorporate these parameters into Eq. (9) to create a toy model of a regularly precessing waveform.

## III. A NEW MODEL OF REGULARLY PRECESSING WAVEFORMS

### A. Defining the new model using precession parameters

We are now ready to construct our regularly precessing waveform. We need to specify the direction of the orbital angular momentum $\hat{\mathbf{L}}$ as a function of GW frequency $f$ and our precession parameters $\langle\theta_{LJ}\rangle$ and $\langle\Omega_{LJ}\rangle$. Once this is determined, the expression $\hat{\mathbf{L}}(f)$ can be inserted into the polarization angle $\psi$ of Eq. (3), the GW amplitude $B$ of Eq. (10), and the GW phase $2\delta\Phi - \phi_p$ of Eqs. (11) and (12) to obtain the waveform $\tilde{h}(f)$. The direction of the orbital angular momentum is given by

$$\hat{\mathbf{L}} = \sin\theta_{LJ}[\cos\Phi_{LJ}\hat{\mathbf{X}}_S + \sin\Phi_{LJ}\hat{\mathbf{Y}}_S] + \cos\theta_{LJ}\hat{\mathbf{Z}}_S\,, \tag{17}$$

where the *source frame* $\{\hat{\mathbf{X}}_S, \hat{\mathbf{Y}}_S, \hat{\mathbf{Z}}_S\}$ is defined such that $\hat{\mathbf{Z}}_S = \hat{\mathbf{J}}$ and $\hat{\mathbf{X}}_S$ is along the line of ascending node of the plane perpendicular to $\hat{\mathbf{J}}$ with respect to the plane of the sky. This frame is depicted in Fig. 1 and described in greater detail in Appendix A.

The precession parameters $\langle\theta_{LJ}\rangle$ and $\langle\Omega_{LJ}\rangle$ defined in the previous section are only constant on the precession timescale, but the angles $\theta_{LJ}$ and $\Phi_{LJ}$ must be specified on the radiation-reaction timescale as the BBH inspirals through the sensitivity band of the GW detector. For simplicity, we assume that these parameter maintain their lowest PN order frequency dependence as given by Eqs. (9) and (10) of [38] all the way to merger,

$$\langle\theta_{LJ}\rangle = 0.1\tilde{\theta}\left(\frac{f}{f_{\rm cut}}\right)^{1/3}, \tag{18a}$$

$$\langle\Omega_{LJ}\rangle = 10^3\,{\rm Hz}\,\tilde{\Omega}\left(\frac{f}{f_{\rm cut}}\right)^{5/3}\left(\frac{M}{M_\odot}\right)^{-1}, \tag{18b}$$

where the normalizations were chosen such that the dimensionless precession amplitude $\tilde{\theta}$ and dimensionless precession frequency $\tilde{\Omega}$ are constants of order unity. We directly equate $\theta_{LJ}$ in Eq. (17) with $\langle\theta_{LJ}\rangle$ in Eq. (18a), while $\Phi_{LJ}$ is obtained by integrating $\langle\Omega_{LJ}\rangle$ in Eq. (18b),

$$\Phi_{LJ} = \gamma_p + \int_{f_{\rm min}}^{f} \langle\Omega_{LJ}\rangle\left(\frac{df'}{dt}\right)^{-1} df'\,, \tag{19}$$

where $f_{\rm min} = 20\,{\rm Hz}$ is the floor of the LVK sensitivity band, the integration constant $\gamma_p$ is the value of $\Phi_{LJ}$ when the binary enters this sensitivity band, and

$$\frac{df}{dt} = \frac{96}{5}\pi\eta f^2 u^5\left[1 - \left(\frac{743}{336} + \frac{11}{4}\eta\right)u^2 + 4\pi u^3\right] \tag{20}$$

is the rate at which the GW frequency increases during the inspiral at 1.5 PN order [26].

MK: Fig. 2 depicts the precession of $\hat{\mathbf{L}}$ according to Eqs. (17) - (19) in the detector frame for equal-mass BBHs with chirp mass $\mathcal{M} = 10M_\odot$. As the GW frequency $f$ increases from $f_{\rm min} = 20\,{\rm Hz}$ to $f_{\rm cut} = 190\,{\rm Hz}$, $\hat{\mathbf{L}}$ spirals in a cone about the direction of the total angular momentum $\hat{\mathbf{J}}$ with increasing opening angle. The left, middle, and right panels in the top row of this figure show three different directions $\hat{\mathbf{J}}$ specified in the detector frame in Table I. Each panels shows three choices of the dimensionless precession amplitude $\tilde{\theta}$ approximately equal to the $5^{\rm th}$, $50^{\rm th}$, and

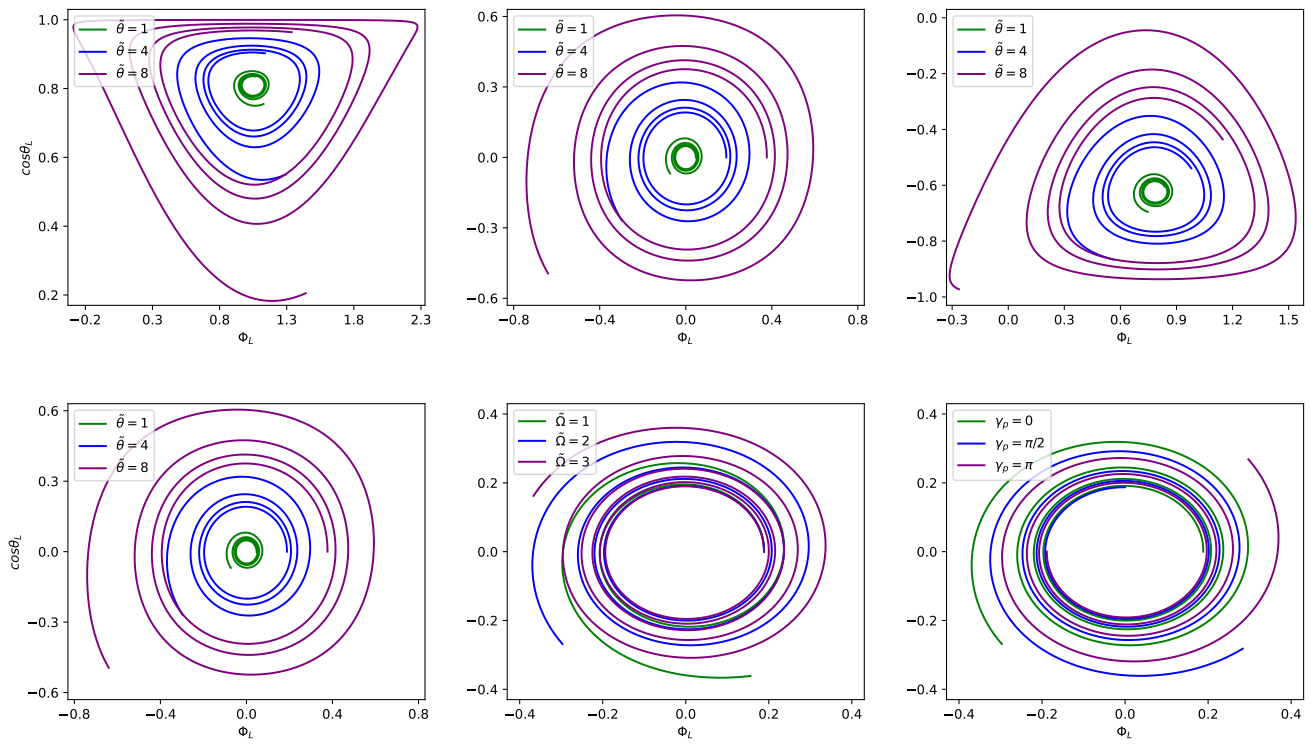



FIG. 2. Precession of the spherical coordinates $\theta_L$ and $\Phi_L$ that specify the direction of orbital angular momentum $\mathbf{L}$ with respect to the detector frame. The BBHs have equal masses and chirp mass $\mathcal{M} = 20M_\odot$. The top row shows this precession for dimensionlesss precession frequency $\tilde{\Omega} = 2$, azimuthal integration constant $\gamma_p = 0$, and dimensionless precession amplitude $\tilde{\theta} = 1$ (green), 4 (blue), and 8 (purple). The left, middle, and right panels of the top row correspond to different directions of the total angular momentum $\mathbf{J}$ listed as Systems 1, 2, and 3 in Table I. The bottom row show the precession of $\mathbf{L}$ for System 2. The left panel is a copy of the middle panel of the top row, the middle panel has $\tilde{\theta} = 4$, $\gamma_p = 0$, and $\tilde{\Omega} = 1$ (green), 2 (blue), and 3 (purple), and the bottom panel has $\tilde{\theta} = 4$, $\tilde{\Omega} = 2$, and $\gamma_p = 0$ (green), $\pi/2$ (blue), and $\pi$ (purple).

95$^{\text{th}}$ percentiles of its distribution for equal-mass, maximally spinning BBHs which are respectively $\tilde{\theta} = 0.95$, 4.23, and 7.75. In the left, middle, and bottom panels of the bottom row of this figure, the direction $\hat{\mathbf{J}}$ remains fixed to that of System 2, but the dimensionless precession amplitude $\tilde{\theta}$, dimensionless precession frequency $\tilde{\Omega}$, and azimuthal integration constant $\gamma_p$ are successively varied. The three choices of $\tilde{\Omega}$ are approximately equal to the 5$^{\text{th}}$, 50$^{\text{th}}$, and 95$^{\text{th}}$ percentiles of its distribution for equal-mass, maximally spinning BBHs which are respectively $\tilde{\Omega} = 1.58$, 2.12, and 2.40, while $\gamma_p$ is expected to have a flat distribution. The effects of these changes on the precession of $\hat{\mathbf{L}}$ are readily apparent; $\tilde{\theta}$ is proportional to the opening angle of the cone, $\tilde{\Omega}$ is proportional to the total angle precessed about the cone during the inspiral, and $\gamma_p$ corresponds to a global rotation of the system about $\hat{\mathbf{J}}$.

MK: Fig. 3 shows how the the GW amplitude $B$ given by Eq. (10) depends on the dimensionless precession amplitude $\tilde{\theta}$, dimensionless precession frequency $\tilde{\Omega}$, and azimuthal integration constant $\gamma_p$ for equal-mass BBHs with chirp mass $\mathcal{M} = 10M_\odot$ and sky location $\theta_S = \pi/4, \Phi_S = 0$. This sky location implies a beam-pattern amplitude $C = 3/4$ and phase $\alpha = 0$ according to Eq. (4).

| Parameter | System 1 | System 2 | System 3 |
|---|---|---|---|
| $\theta_J$ | $\pi/4$ | $\pi/2$ | $8\pi/9$ |
| $\Phi_J$ | 0 | $\pi/2$ | $\pi/4$ |
| $\cos\iota_{JN}$ | 1 | 0 | -0.493 |
| $\psi_0$ | — | 0 | -1.29 |

TABLE I. Spherical coordinates $\theta_J, \Phi_J$ of the total angular momentum $\mathbf{J}$ in the detector frame, inclination $\iota_{JN}$ between $\hat{\mathbf{J}}$ and the sky location $\hat{\mathbf{N}}$, and polarization angle $\psi_0$ for zero precession amplitude. All three BBH systems have equal-mass BBHs with chirp mass $\mathcal{M} = 10M_\odot$ and sky location $\theta_S = \pi/4, \Phi_S = 0$.

A necessary but insufficient condition for precession to be detected with confidence is for the number of precession

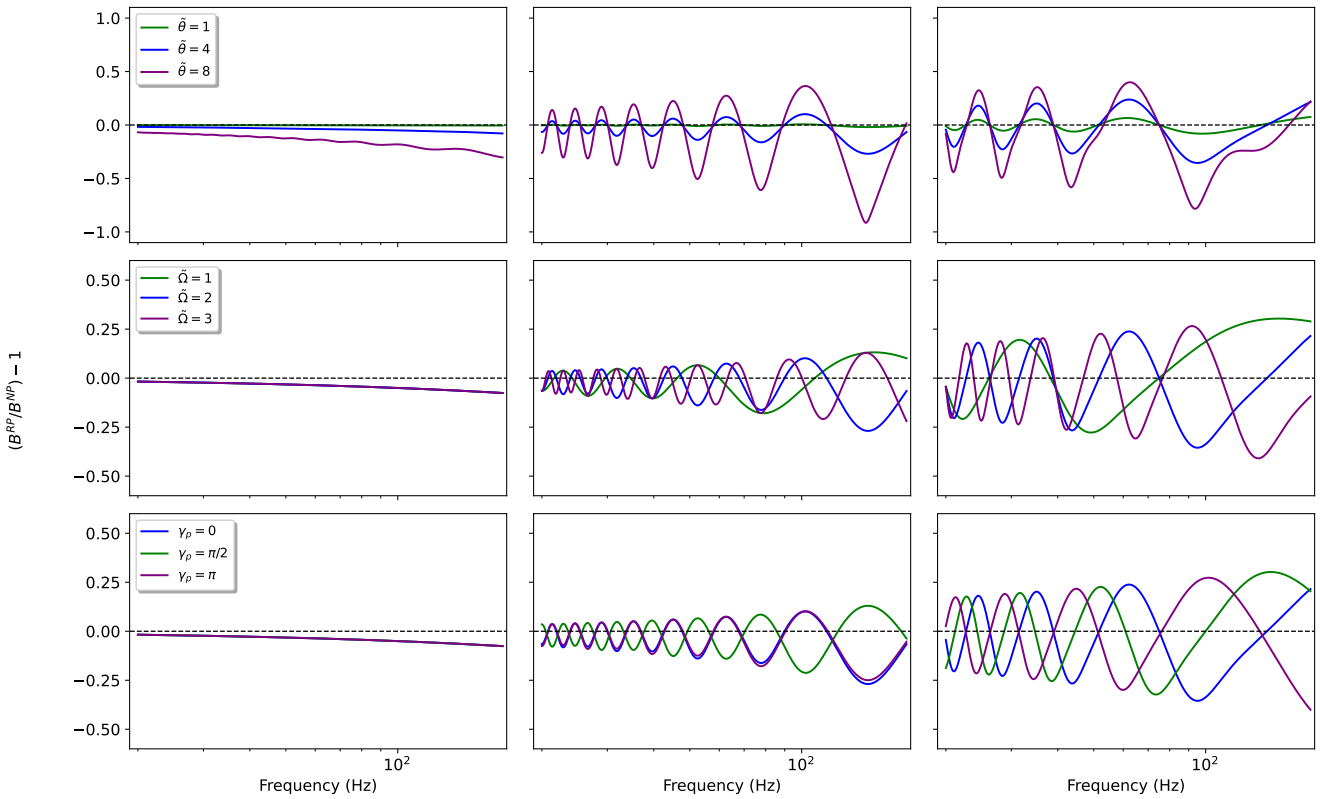



FIG. 3. MK: The ratio of the GW amplitude $B$ given by Eq. (10) in our regularly precessing model (RP) to a non-precessing (NP) model with the same BBH masses. The left, middle, and right panels correspond to directions of the total angular momentum $\hat{\mathbf{J}}$ listed as Systems 1, 2, and 3 in Table I. The top panels fix $\tilde{\Omega} = 2$, $\gamma_p = 0$ and vary $\tilde{\theta}$, the middle panels fix $\tilde{\theta} = 4$, $\gamma_p = 0$ and vary $\tilde{\Omega}$, and the bottom panels fix $\tilde{\theta} = 4$, $\tilde{\Omega} = 2$ and vary $\gamma_p$.

cycles

$$N_{\rm pre} = \frac{1}{2\pi}[\Phi_{LJ}(f_{\rm cut}) - \Phi_{LJ}(f_{\rm min})] \qquad (21)$$

occurring within the sensitivity band of the detector to be greater than unity. In Fig. 4, the dot-dashed, dashed, and dotted horizontal lines show the $5^{\rm th}$, $50^{\rm th}$, and $95^{\rm th}$ percentiles of the distribution of the dimensionless precession frequency $\tilde{\Omega}$ for maximal spins (left panel) and equal-mass binaries (right panel). We see that the median values of $\tilde{\Omega}$ are weakly sensitive to both the mass ratio $q$ and spin magnitudes $\chi_i$, and that dispersion in $\tilde{\Omega}$ increases with $\chi_i$ for equal-mass BBHs. This is consistent with the behavior of $\langle \Omega_{LJ} \rangle$ at $r = 10M$ ($u = 10^{-1/2}$) shown in the middle right panel of Fig. 3 of [38]. The solid curves show the values of $\tilde{\Omega}$ for which $N_{\rm pre} = 1$. The portions of parameter space above these curves are more likely to exhibit detectable precession. We see that, for a wide range of mass ratios and spins, a majority of binaries satisfy the criterion $N_{\rm pre} \geq 1$ for $M \lesssim 45 M_\odot$, and that 5% of the distribution could have detectable precession for $q \geq 0.5$, $\chi_i \approx 1$, and $M \lesssim 60 M_\odot$.

The dependence of the amplitude $B(f)$ on our three new parameters is shown in Fig. 3. We choose equal-mass BBHs with chirp mass $\mathcal{M} = 20 M_\odot$ with a sky location $\theta_S = \pi/4$, $\Phi_S = 0$. The left, middle, and right columns correspond to three different choices of the direction of the total angular momentum $\mathbf{J}$ listed as Systems 1, 2, and 3 in Table I. Careful examination of Eq. (10) reveals that the amplitude depends linearly on $A$, independently up to $f^{7/6}$, on the beam pattern coefficients and the inner product $\hat{\mathbf{L}} \cdot \hat{\mathbf{N}}$. We plot the ratio of the RP over the NP amplitude, in order to focus on the effect of our precession parameters on the amplitude and to demonstrate the aforementioned dependencies more clearly.

We derive a lot of useful conclusions by closely examining the behavior of the various ratios from Fig. (3). ES:rewrite the sentence.. We have previously shown that the number of precession cycles in band depends linearly on $\tilde{\Omega}$. The middle row of Fig. 3 shows that increasing $\tilde{\Omega}$ also increases the number of precession cycles, while leaving the precession amplitude intact. $\gamma_p$ as defined in Eq. (19) is one of the three parameters that emerge when we include precession effects. It expresses the opening azimuthal angle between $\hat{\mathbf{L}}$ and $\hat{\mathbf{J}}$ in the orbital frame, when $\hat{\mathbf{L}}$ enters the frequency band of the detector. Therefore, we are going to refer to $\gamma_p$ as a *'nuisance parameter'*. As expected, the bottom row of Fig. 3 reveals that varying $\gamma_p$ only shifts the amplitude ratios to the left.

To acquire a better understanding of these cases, we

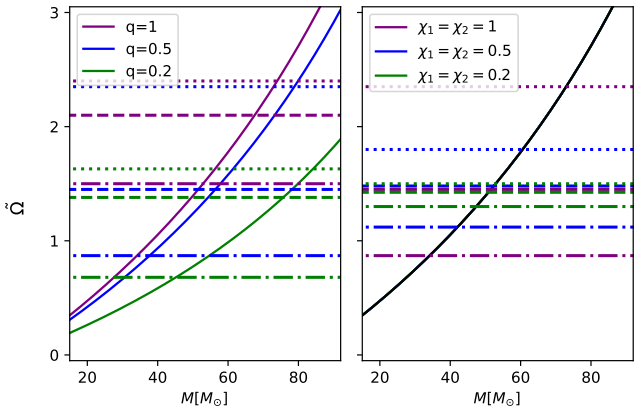



FIG. 4. Dimensionless precession frequency $\tilde{\Omega}$ as a function of total binary mass $M$. In the left panel, the dot-dashed, dashed, and dotted horizontal lines show the $5^{\text{th}}$, $50^{\text{th}}$, and $95^{\text{th}}$ percentiles of the $\tilde{\Omega}$ distribution for BBHs with isotropic maximal spins and mass ratios $q = 1$ (purple), $q = 0.5$ (blue), and $q = 0.2$ (green). In the right panel, the dot-dashed, dashed, and dotted horizontal lines show the $5^{\text{th}}$, $50^{\text{th}}$, and $95^{\text{th}}$ percentiles of the $\tilde{\Omega}$ distribution for equal-mass BBHs with dimensionless spins magnitudes $\chi_i = 1$ (purple), $\chi_i = 0.5$ (blue), and $\chi_i = 0.2$ (green). The solid curves in both panels show the minimum value of $\tilde{\Omega}$ for which at least one precession cycle occurs after the BBHs enter the LVK sensitivity band at a GW frequency of 20 Hz.

project $\hat{\mathbf{L}}$ in the detector frame

$$\hat{\mathbf{L}} = \sin\theta_L(\cos\Phi_L\hat{\mathbf{X}}_D + \sin\Phi_L\hat{\mathbf{Y}}_D) + \cos\theta_L\hat{\mathbf{Z}}_D\,. \quad (22)$$

It follows, that when we are in the NP regime $\theta_L \equiv \theta_J$, $\Phi_J \equiv \Phi_L$ and $\hat{\mathbf{L}}\cdot\hat{\mathbf{N}}$ is simply given by Eq. (A4). Hence, it is straightforward to plot contour plots for an isotropic distribution of $\theta_L, \Phi_L$ for the three terms that describe the amplitude of the waveform in Eq. (10). In addition, we can infer that by including precession effects, $\theta_L, \Phi_L$ become dependent on our three precession parameters as well as the frequency. ES:connect it with the appendixLooking back at Eq. (22), we have:

$$\cos\theta_L(f) = \hat{\mathbf{L}}\cdot\hat{\mathbf{Z}}_D \quad (23)$$

$$\tan\Phi_L(f) = \frac{\hat{\mathbf{L}}\cdot\hat{\mathbf{Y}}_D}{\hat{\mathbf{L}}\cdot\hat{\mathbf{X}}_D}\,, \quad (24)$$

where we can calculate these angles by combining Eq. (22) with Eqs. (**??**)-(**??**). ES:needs revising to explain the 2 new figures

Other cases where the terms in our GW signal can vanish are the *face-on* and *edge-on* cases. For example, we call a BBH system face-on when the angle between the total orbital angular momentum and the line of sight is $\theta_{JN} = 0$ and edge-on when $\theta_{JN} = \pi/2$ or in our case, when $\iota_{JN} = 0$ and $\pi/2$ respectively. For the face-on NP case we know that $\hat{\mathbf{J}} \parallel \hat{\mathbf{L}}$. So, we get $\hat{\mathbf{L}}\cdot\hat{\mathbf{N}} = 1$ and the cross polarization term in Eq. (5b) vanishes. On top of that, for a particular choice of the sky localization angles in the NP limit, we can make either $\mathrm{F}_+ = 0$ or $\mathrm{F}_\times = 0$. Having either of these terms equal to 0 affects the calculation of the amplitude both in the NP and the RP regime and that is one of the key factors that lead to these behaviors in Fig. 3 as we have already discussed. It also implies that the location of the BBH system in the sky, can in fact make either one of the terms that describe the GW polarizations vanish. This has already been well studied and stated in the literature, i.e. [46]. It was shown that, we are observationally biased towards detecting binaries, in which the orbital angular momentum is either aligned or antialigned with the line of sight.

Fig. (**??**) presents an example of a face-on and edge-on case. There exist infinite combinations of the four angles that specify the sky localization of the BBH, that can give us a face-on and an edge-on case respectively. In our example, we have picked the simplest cases for the angle choices, that is for the face-on case we have set $(\theta_S, \theta_J) = (\pi/6, \pi/6)$ and we have chosen a random value for the azimuthal angles seeing as the terms that depend on these two angles in Eq. (A4) vanish. As for the edge-on case, we choose $(\theta_S, \theta_J) = (\pi/6, 2\pi/3)$ and we set the azimuthal angles equal to each other, $\Phi_S = \Phi_J = \pi/3$. Taking a closer look at the left panel of Fig. (**??**), we find out that the face-on case confirms our analysis. Even though, we have introduced RP through non-zero values of our precession parameters, the term $\sin\iota_{JN} = 0$ forces the term that gives us the periodic precessing behavior in Eq. (A10) to vanish. In turn, this transforms the ratios into ES:straight lines with different slopes depending on the value of the precession amplitude.

Things get more interesting when we move over to the edge-on case and the right panel of Fig. (**??**). By closer examination, we understand that even though we have used the same value of $\tilde{\Omega}$, the number of precession cycles in band almost gets doubled. Going back to Eq. (A10), we note that for the edge-on case the second term is equal to 0. That means, when we square the $\hat{\mathbf{L}}\cdot\hat{\mathbf{N}}$ term in order to calculate Eq. (10), we only get a single term.....

We have shown previously that by introducing RP and in turn frequency dependence to the inclination angle between the orbital angular momentum and the line of sight, the phase of the GW template gets modulated via two terms $\phi_p$ and $\delta\Phi$ respectively. Figs. 5 and 6 show the RP behavior of these terms for different choices of sky localization. Once again, in both terms but more noticeably in the polarization angle $\phi_p$, we see that various choice of the angles produce different behaviors. By looking at Eqs. (11) and (12), it is again evident that the values of the beam pattern coefficients and more importantly $\hat{\mathbf{L}}\cdot\hat{\mathbf{N}}$ affect the phase correction terms in various degrees. However, we find it important to note the one big difference of these two phase terms. The $\phi_p$ term is calculated from Eq. (11) as the inverse tangent of the imaginary over the real part of the waveform, so its value follows a periodic behavior. ES: explain this–what Mike has been saying secular etc.

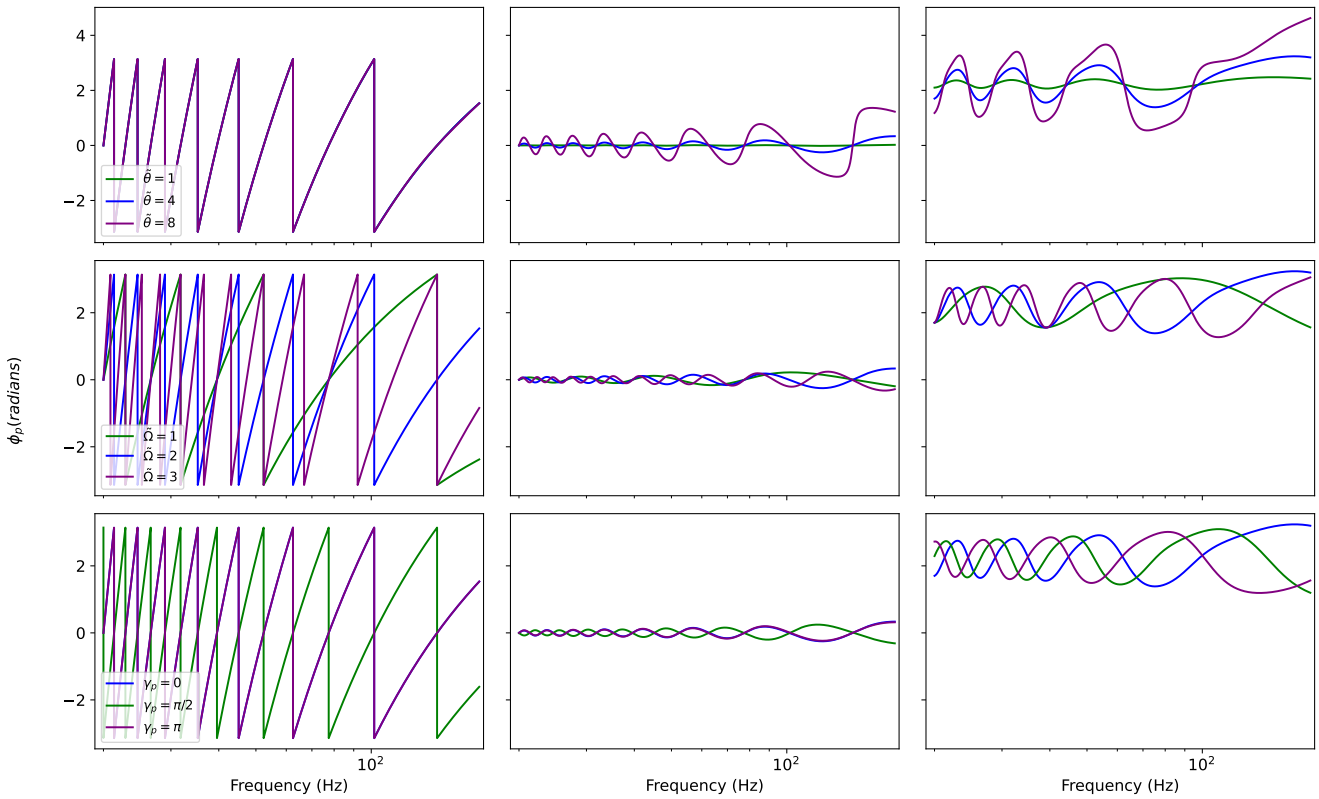



FIG. 5. The evolution of the polarization phase $\phi_p$ given by Eq. (11) in our regularly precessing model (RP) with respect to the frequency. The left, middle, and right panels correspond to directions of the total angular momentum $\hat{\mathbf{J}}$ listed as Systems 1, 2, and 3 in Table I. The top panels fix $\tilde{\Omega}=2$, $\gamma_p=0$ and vary $\tilde{\theta}$, the middle panels fix $\tilde{\theta}=4$, $\gamma_p=0$ and vary $\tilde{\Omega}$, and the bottom panels fix $\tilde{\theta}=4$, $\tilde{\Omega}=2$ and vary $\gamma_p$.

## IV. SIGNAL-TO-NOISE AND MISMATCH CALCULATIONS

In this section, we will employ the matched-filtering tools on our new RP 'toy' model and arrive to several useful results and conclusions. An inner product on the parameter vector space between two signals expressed in the frequency domain, $\tilde{h}(f)$, $\tilde{g}(f)$, is given by [47, 48]:

$$\langle h|g\rangle = 4\,\mathrm{Re}\int_{f_{\min}}^{f_{\rm cut}} \frac{h(f)g^*(f)}{S_n(f)}df\,, \tag{25}$$

where $S_n(f)$ is the power spectral density (PSD) of the detector noise.

We can then use Eq (25) to compute the signal-to-noise (SNR). The SNR provides a measure of the strength of the GW signal arriving on the detector compared to the background noise. We can obtain the SNR between a waveform signal arriving from the source, $h_s$, and a waveform signal constructed from a template, $h_t$, [26]:

$$\rho = \langle h_s|h_t\rangle^{1/2} \tag{26}$$

We also define the overlap between the two signals from [26] such that:

$$\mathcal{O}(h_s,h_t) = \frac{\langle h_s|h_t\rangle}{\sqrt{\langle h_s|h_s\rangle\langle h_t|h_t\rangle}}\,, \tag{27}$$

where the overlap is constrained to be $\mathcal{O}\leq 1$.

The *match* is defined as the overlap maximized over the time and the phase of coalescence:

$$\mathrm{M}(\mathrm{h_s},\mathrm{h_t}) = \max_{\mathrm{t_c},\phi_c}\mathcal{O}(\mathrm{h_s},\mathrm{h_t})\,, \tag{28}$$

while the mismatch $\epsilon$ is just:

$$\epsilon(h_s,h_t) = 1-\mathrm{M}\,, \tag{29}$$

where we use the PYCBC.FILTER package [49] to evaluate the mismatch between waveforms and to pick various PSD.

Fig. 7 shows plots of the mismatch $\epsilon$ between a RP GW source and a NP template used by a ground based L-shaped detector as a function of the precession amplitude $\tilde{\theta}$(left panels) and the precession frequency $\tilde{\Omega}$(right panels) for a chirp mass of $10M_\odot$(top panels) and $20M_\odot$(bottom panels). We use the same source parameters for the RP and the NP templates from System II in Table I and the noise PSD appropriate for a single two-armed

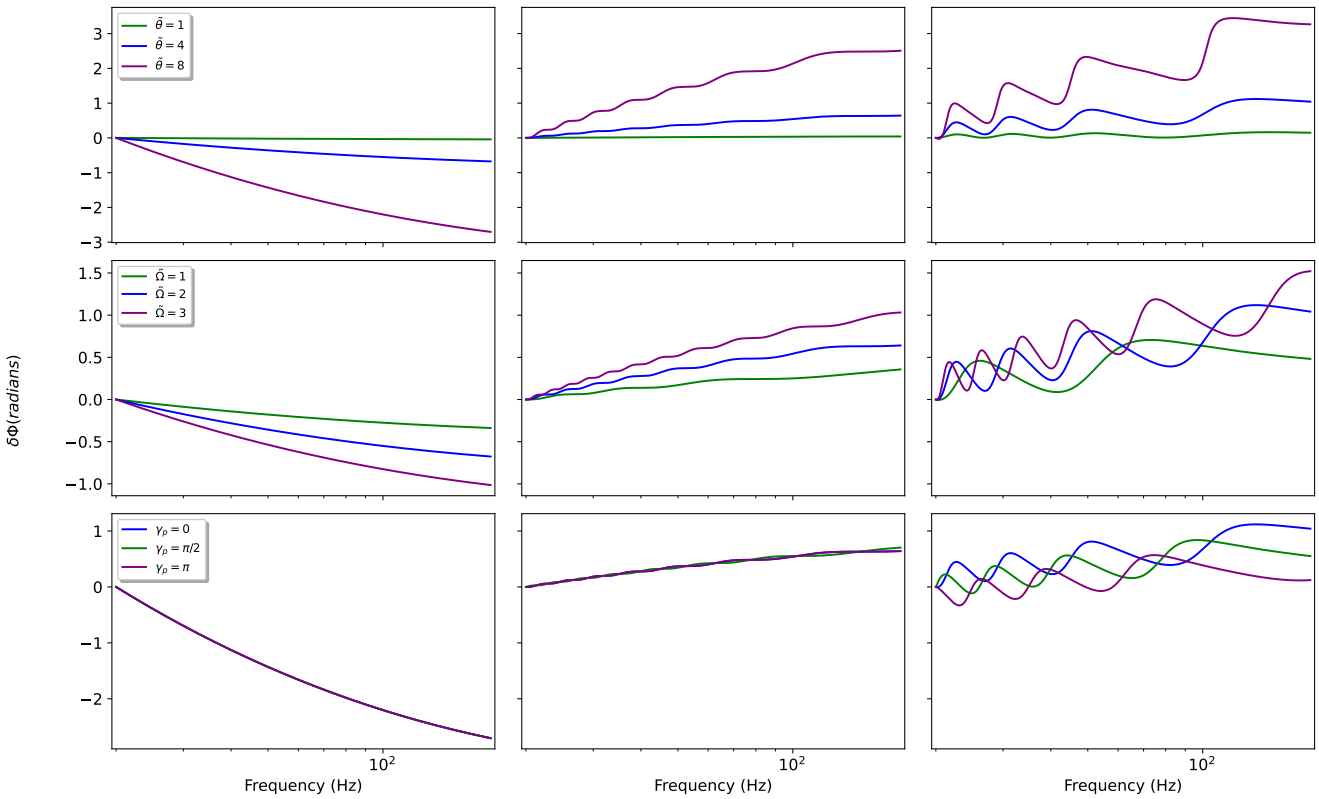



FIG. 6. The evolution of the phase correction term $\delta\Phi$ given by Eq. (12) in our regularly precessing model (RP) with respect to the frequency. The left, middle, and right panels correspond to directions of the total angular momentum $\hat{\mathbf{J}}$ listed as Systems 1, 2, and 3 in Table I. The top panels fix $\tilde{\Omega} = 2$, $\gamma_p = 0$ and vary $\tilde{\theta}$, the middle panels fix $\tilde{\theta} = 4$, $\gamma_p = 0$ and vary $\tilde{\Omega}$, and the bottom panels fix $\tilde{\theta} = 4$, $\tilde{\Omega} = 2$ and vary $\gamma_p$.

detector with aLIGO design sensitivity. In addition, when we calculate $\epsilon$ for the left column, we pick $\tilde{\Omega} = 1, 2$ corresponding to the 5% and 50% confidence levels, while we follow the same method for the right panels and we calculate $\epsilon$ for two different values of $\tilde{\theta}$. Fig. 7 confirms that $\epsilon \to 0$, when we have either $\tilde{\theta} \to 0$ or $\tilde{\Omega} \to 0$.

Moreover, we understand that by increasing the chirp mass of the BBH system we are decreasing the upper limit, $f_{\rm ISCO}$, of the integral calculations in Eq. (25). That implies that the BBH system is spending less time in the observable frequency band of the detector and that in turn is decreasing the number of precession cycles we are going to detect for the same value of $\tilde{\Omega}$. By doubling the chirp mass between calculations we are almost cutting the $f_{\rm ISCO}$ in half. This assumption is confirmed when we take a look at the maximum values of the mismatches in each case. The other main difference between the two columns of Fig. 7 is the dependence of $\epsilon$ on the two of our three RP parameters. For the left panels of Fig. 7, the evolution of $\epsilon$ with respect to $\tilde{\theta}$ for constant $\tilde{\Omega}$ is fairly easy to interpret. However, the behavior of $\epsilon$ in the right panels presents features that are difficult to interpret. The main reason for that effect is the number of precession cycles that is monotonically increasing when we increase $\tilde{\Omega}$. We present Fig. 8 in an attempt to clarify this complicated behavior.

## V. DISCUSSION

It is widely accepted that when the individual spins of the BHs that comprise a BBH system are non-zero and misaligned with one another and the angular momentum $\hat{\mathbf{L}}$, they can cause the orbital plane to precess and nutate. These spin-orbit and spin-spin coupling effects are the reason behind the introduction of modulations to the GW template's amplitude and phase at various PN orders. These modulations have to be modelled as accurately as possible in order to extract the necessary information concerning the parameters that describe the BBH system in the current LVK runs as well as the next 3G detectors(from statement). In our previous study [38], we have introduced a taxonomy that incorporates all the different cases that can cause precession and nutation to happen and we also came up with 5 novel phenomenological parameters in the inspiral PN regime.

In this paper, We propose a new spin-independent method to model the precessional effects on the GW by constructing a new 'toy' model waveform and introducing 3 new parameters in the RP limit, namely $\theta_{LJ}$, $\Phi_{LJ}$ and

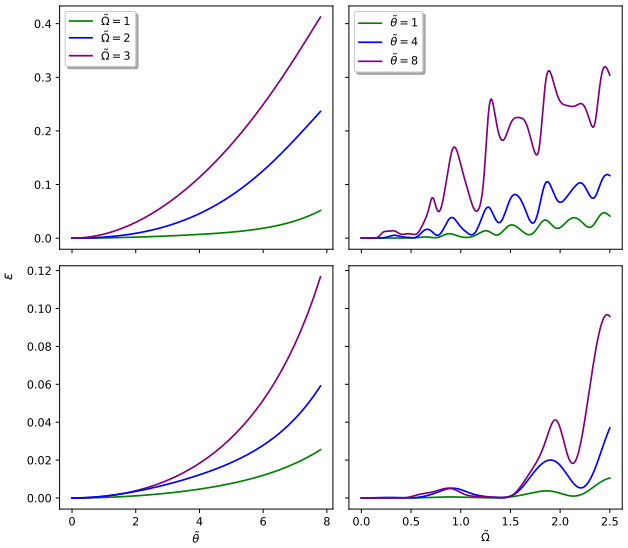



FIG. 7. The evolution of the mismatch $\epsilon$ given by Eq. (29) in our regularly precessing model (RP) with respect to the normalized precession amplitude $\tilde{\theta}$(left panels) and the normalized precession frequency $\tilde{\Omega}$(right panels). All four panels correspond to the direction of the total angular momentum $\hat{\mathbf{J}}$ listed as System 2 in Table I and the direction of the line of sight $\hat{\mathbf{N}}$ is specified by $(\theta_S, \Phi_S) = (\pi/4, 0)$ . The values of the chirp mass $\mathcal{M}$ are $\mathcal{M} = 10M_\odot$(top panels) and $\mathcal{M} = 20M_\odot$(bottom panels). The panels have a fixed $\gamma_p = 0$ and varying $\tilde{\Omega}$(left panels) and $\tilde{\theta}$(right panels).

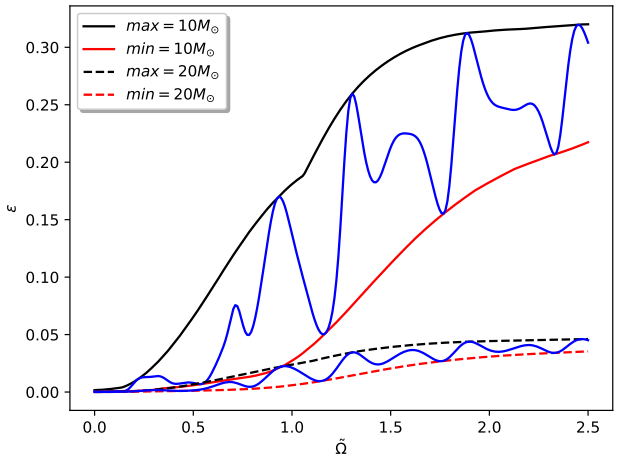



FIG. 8. Mismatches between a non-precessing GW source and our RP template as a function of the precession frequency $\tilde{\Omega}$. We then vary $\gamma_p$ from 0 to 360 for each value of $\tilde{\Omega}$ and find the maximum and minimum value of the mismatch independently. The black(red) solid line is showing the maximum(minimum) envelope for the mismatch values of each individual $\gamma_p$ for $\mathcal{M} = 10M_\odot$, while the dashed lines are calculated for $\mathcal{M} = 20M_\odot$.ES:change the colors

$\gamma_p$. In turn, We have shown that these parameters come inside the terms given by Eqs. ()() and most importantly into $LN$, making all of these terms depend not only on the frequency but also on two dimensionless RP pameterers, $\tilde{\theta}$, $\tilde{\Omega}$ as well as $\gamma_p$. We illustrate in Fig. 3 these parameters affect independently the precession amplitude, the precession frequency and the initial phase? ES:its shit.

2nd In this paper - we have tried to simulate the LVK results by building a new precessing toy model being described by the normal parameters 8(localization,dist,chirp) and 3 new precession parameters- We described LN through our new parameters -introduced a frequency dependecy to LN and in turn to the three terms that fully describe a waveform template amplitude, phip, deltaphi- we have shown that the first parameter in FIGS change only the amplitude- the second only the frequency and the number of cycles in band and the third is just a nuisance parameter.

3rd SNR mismatches strange effects FIG8

4th talk about 3G detectors will help discover precession even with toy model templates like ours - say something about a system of detectors? - next step is already being done by making a toy model template that described microlensing effects - combination paper how easy it is to distinguish between these two effects - maybe in the future a precessing event can be shown as a lensing event and vice versa - stopped at RP next step taking the full range of phenomenological parameters and building a fully functional waveform family comparable to IMPHenom(maybe higher harmonics-comparisons between the two). 5th astro para from Nate. NAS: Astro implications para.

## ACKNOWLEDGMENTS

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## Appendix A: Three reference frames

We use three distinct reference frames to calculate our regularly precessing waveforms.

### 1. Detector frame

The *detector frame* is chosen such that its basis vectors $\hat{\mathbf{X}}_D$ and $\hat{\mathbf{Y}}_D$ point along the arms of the interferometer, and $\hat{\mathbf{Z}}_D = \hat{\mathbf{X}}_D \times \hat{\mathbf{Y}}_D$ points towards the zenith at the detector location. The sky location of the binary $\hat{\mathbf{N}}$ and the direction of the total angular momentum $\hat{\mathbf{J}}$ are

specified respectively by the angles $(\theta_S, \Phi_S)$ and $(\theta_J, \Phi_J)$ with respect to this frame:

$$\hat{\mathbf{N}} = \sin\theta_S(\cos\Phi_S\hat{\mathbf{X}}_D + \sin\Phi_S\hat{\mathbf{Y}}_D) + \cos\theta_S\hat{\mathbf{Z}}_D\,, \quad \text{(A1a)}$$

$$\hat{\mathbf{J}} = \sin\theta_J(\cos\Phi_J\hat{\mathbf{X}}_D + \sin\Phi_J\hat{\mathbf{Y}}_D) + \cos\theta_J\hat{\mathbf{Z}}_D\,. \quad \text{(A1b)}$$

### 2. Sky frame

The *sky frame* is chosen such that $\hat{\mathbf{N}}$ points into the plane of the sky (opposite to the direction of GW propagation), $\hat{\mathbf{V}}$ points towards the zenith, and $\hat{\mathbf{H}} = \hat{\mathbf{V}} \times \hat{\mathbf{N}}$ is parallel to the horizon:

$$\hat{\mathbf{H}} = -\sin\Phi_S\hat{\mathbf{X}}_D + \cos\Phi_S\hat{\mathbf{Y}}_D\,, \quad \text{(A2a)}$$

$$\hat{\mathbf{V}} = -\cos\theta_S(\cos\Phi_S\hat{\mathbf{X}}_D + \sin\Phi_S\hat{\mathbf{Y}}_D) + \sin\theta_S\hat{\mathbf{Z}}_D\,. \quad \text{(A2b)}$$

### 3. Source frame

The *source frame* is chosen such that $\hat{\mathbf{Z}}_S$ points along the direction of the total angular momentum $\hat{\mathbf{J}}$,

$$\hat{\mathbf{X}}_S = \frac{\hat{\mathbf{N}} \times \hat{\mathbf{J}}}{|\hat{\mathbf{N}} \times \hat{\mathbf{J}}|} \quad \text{(A3)}$$

points along the line of ascending node of the plane perpendicular to $\mathbf{J}$ with respect to the sky plane spanned by $\hat{\mathbf{H}}$ and $\hat{\mathbf{V}}$, and $\hat{\mathbf{Y}}_S = \hat{\mathbf{Z}}_S \times \hat{\mathbf{X}}_S$. We define the inclination $\iota_{JN}$ as the angle between $\hat{\mathbf{J}}$ and $\hat{\mathbf{N}}$,

$$\begin{aligned}\cos\iota_{JN} &= \hat{\mathbf{J}} \cdot \hat{\mathbf{N}} \quad \text{(A4)}\\ &= \sin\theta_J\sin\theta_S\cos(\Phi_J - \Phi_S) + \cos\theta_J\cos\theta_S\,,\end{aligned}$$

and the longitude of ascending node $\Omega_{XH}$ such that the source and sky frames are related by

$$\hat{\mathbf{X}}_S = \cos\Omega_{XH}\hat{\mathbf{H}} + \sin\Omega_{XH}\hat{\mathbf{V}}\,, \quad \text{(A5a)}$$

$$\hat{\mathbf{Y}}_S = \cos\iota_{JN}(-\sin\Omega_{XH}\hat{\mathbf{H}} + \cos\Omega_{XH}\hat{\mathbf{V}}) + \sin\iota_{JN}\hat{\mathbf{N}}\,, \quad \text{(A5b)}$$

$$\hat{\mathbf{Z}}_S = \sin\iota_{JN}(\sin\Omega_{XH}\hat{\mathbf{H}} - \cos\Omega_{XH}\hat{\mathbf{V}}) + \cos\iota_{JN}\hat{\mathbf{N}}\,, \quad \text{(A5c)}$$

where

$$\sin\Omega_{XH} = \frac{\sin\theta_J\sin(\Phi_J - \Phi_S)}{\sin\iota_{JN}}\,, \quad \text{(A6a)}$$

$$\cos\Omega_{XH} = \frac{\sin\theta_J\cos\theta_S\cos(\Phi_J - \Phi_S) - \cos\theta_J\sin\Phi_S}{\sin\iota_{JN}}\,. \quad \text{(A6b)}$$

In the aligned case ($\iota_{JN} = 0$), Eq. (A6) breaks down and one can choose $\Omega_{XH} = 0$ without loss of generality (it is degenerate with the integration constant $\gamma_P$).

In the source and sky frames, the direction of the orbital angular momentum $\hat{\mathbf{L}}$ is given by

$$\begin{aligned}\hat{\mathbf{L}} &= \sin\theta_{LJ}[\cos\Phi_{LJ}\hat{\mathbf{X}}_S + \sin\Phi_{LJ}\hat{\mathbf{Y}}_S] + \cos\theta_{LJ}\hat{\mathbf{Z}}_S\,, \quad \text{(A7)}\\ &= [\sin\theta_{LJ}(\cos\Phi_{LJ}\cos\Omega_{XH} - \sin\Phi_{LJ}\cos\iota_{JN}\sin\Omega_{XH}) + \sin\iota_{JN}\sin\Omega_{XH}\cos\theta_{LJ}]\hat{\mathbf{H}}\\ &\quad + [\sin\theta_{LJ}(\cos\Phi_{LJ}\sin\Omega_{XH} + \sin\Phi_{LJ}\cos\iota_{JN}\cos\Omega_{XH}) - \sin\iota_{JN}\cos\Omega_{XH}\cos\theta_{LJ}]\hat{\mathbf{V}}\\ &\quad + (\sin\theta_{LJ}\sin\Phi_{LJ}\sin\iota_{JN} + \cos\theta_{LJ}\cos\iota_{JN})\hat{\mathbf{N}}\,. \quad \text{(A8)}\end{aligned}$$

The source and sky frames are depicted in green and red respectively in Fig. 1. In our regularly precessing waveform model, only the angles $\Phi_{LJ}$ and $\theta_{LJ}$ are time dependent, and they vary on the precession and radiation-reaction timescales respectively according to Eqs. (19) and (18a).

### 4. GW strain

The observed GW strain given by Eq. (1) depends on the relative orientation of these three frames through the GW fields $h_+$ and $h_\times$ of Eq. (5) and the detector beam-pattern coefficients $F_+$ and $F_\times$ of Eq. (2). The latter depend on the polarization angle $\psi$ of Eq. (3) which is given in terms of the above angles as

$$\begin{aligned}\hat{\mathbf{L}}\cdot\hat{\mathbf{Z}}_D &= \sin\theta_{LJ}[\cos\Phi_{LJ}\sin\Omega_{XH}\sin\theta_S + \sin\Phi_{LJ}(\cos\iota_{JN}\cos\Omega_{XH}\sin\theta_S + \sin\iota_{JN}\cos\theta_S)] \\ &\quad + \cos\theta_{LJ}(\cos\iota_{JN}\cos\theta_S - \sin\iota_{JN}\cos\Omega_{XH}\sin\theta_S)\,, \qquad \text{(A9)}\\ \hat{\mathbf{L}}\cdot\hat{\mathbf{N}} &= \sin\theta_{LJ}\sin\iota_{JN}\sin\Phi_{LJ} + \cos\theta_{LJ}\cos\iota_{JN}\,, \qquad \text{(A10)}\\ \hat{\mathbf{N}}\cdot\hat{\mathbf{Z}}_D &= \cos\theta_S\,, \qquad \text{(A11)}\\ \hat{\mathbf{N}}\cdot(\hat{\mathbf{L}}\times\hat{\mathbf{Z}}_D) &= \hat{\mathbf{Z}}_D\cdot(\hat{\mathbf{N}}\times\hat{\mathbf{L}}) \\ &= [\sin\theta_{LJ}(\cos\Phi_{LJ}\cos\Omega_{XH} - \sin\Phi_{LJ}\cos\iota_{JN}\sin\Omega_{XH}) + \sin\iota_{JN}\sin\Omega_{XH}\cos\theta_{LJ}]\sin\theta_S\,, \qquad \text{(A12)}\\ \tan\psi &= \frac{\hat{\mathbf{L}}\cdot\hat{\mathbf{Z}}_D - (\hat{\mathbf{L}}\cdot\hat{\mathbf{N}})(\hat{\mathbf{N}}\cdot\hat{\mathbf{Z}}_D)}{\hat{\mathbf{N}}\cdot(\hat{\mathbf{L}}\times\hat{\mathbf{Z}}_D)} \qquad \text{(A13)}\\ &= \frac{\sin\theta_{LJ}(\cos\Phi_{LJ}\sin\Omega_{XH} + \sin\Phi_{LJ}\cos\iota_{JN}\cos\Omega_{XH}) - \cos\theta_{LJ}\sin\iota_{JN}\cos\Omega_{XH}}{\sin\theta_{LJ}(\cos\Phi_{LJ}\cos\Omega_{XH} - \sin\Phi_{LJ}\cos\iota_{JN}\sin\Omega_{XH}) + \cos\theta_{LJ}\sin\iota_{JN}\sin\Omega_{XH}} \qquad \text{(A14)}\end{aligned}$$

In the aligned case ($\iota_{JN} = 0$), Eq. (A14) above reduces to $\psi = \Phi_{LJ} + \Omega_{XH}$, and the longitude of ascending node $\Omega_{XH}$ is degenerate with the integration constant $\gamma_P$. We can also use these three reference frame to evaluate the contribution $\delta\Phi$ to the GW phase given by Eq. (12):

$$\begin{aligned}\frac{d\hat{\mathbf{L}}}{df} &= \langle\Omega_{LJ}\rangle\left(\frac{df}{dt}\right)^{-1}\sin\theta_{LJ}(-\sin\Phi_{LJ}\hat{\mathbf{X}}_S + \cos\Phi_{LJ}\hat{\mathbf{Y}}_S) \qquad \text{(A15)}\\ (\hat{\mathbf{L}}\times\hat{\mathbf{N}})\cdot\hat{\mathbf{X}}_S &= (\hat{\mathbf{X}}_S\times\hat{\mathbf{L}})\cdot\hat{\mathbf{N}} = (\sin\theta_{LJ}\sin\Phi_{LJ}\hat{\mathbf{Z}}_S - \cos\theta_{LJ}\hat{\mathbf{Y}}_S)\cdot\hat{\mathbf{N}} = \sin\theta_{LJ}\sin\Phi_{LJ}\cos\iota_{JN} - \cos\theta_{LJ}\sin\iota_{JN} \qquad \text{(A16)}\\ (\hat{\mathbf{L}}\times\hat{\mathbf{N}})\cdot\hat{\mathbf{Y}}_S &= (\hat{\mathbf{Y}}_S\times\hat{\mathbf{L}})\cdot\hat{\mathbf{N}} = (-\sin\theta_{LJ}\cos\Phi_{LJ}\hat{\mathbf{Z}}_S + \cos\theta_{LJ}\hat{\mathbf{X}}_S)\cdot\hat{\mathbf{N}} = -\sin\theta_{LJ}\cos\Phi_{LJ}\cos\iota_{JN} \qquad \text{(A17)}\\ \frac{d\delta\Phi}{df} &= \left[\frac{\hat{\mathbf{L}}\cdot\hat{\mathbf{N}}}{1-(\hat{\mathbf{L}}\cdot\hat{\mathbf{N}})^2}\right](\hat{\mathbf{L}}\times\hat{\mathbf{N}})\cdot\frac{d\hat{\mathbf{L}}}{df} \\ \frac{d\delta\Phi}{df} &= \left[\frac{\hat{\mathbf{L}}\cdot\hat{\mathbf{N}}}{1-(\hat{\mathbf{L}}\cdot\hat{\mathbf{N}})^2}\right]\langle\Omega_{LJ}\rangle\left(\frac{df}{dt}\right)^{-1}\sin\theta_{LJ}(\cos\theta_{LJ}\sin\iota_{JN}\sin\Phi_{LJ} - \sin\theta_{LJ}\cos\iota_{JN}) \qquad \text{(A18)}\end{aligned}$$

## Appendix B: Notes: NOT FOR PUBLICATION

Here are some useful formulae:

$$C = \left[\frac{1}{4}(1+\cos\theta_S^2)^2\cos^2 2\Phi_S + \cos^2\theta_S \sin^2 2\Phi_S\right]^{1/2} \tag{B1}$$

$$\sin\alpha = \frac{1}{C}\cos\theta_S \sin 2\Phi_S \tag{B2}$$

$$\cos\alpha = \frac{1}{2C}(1+\cos^2\theta_S)\cos 2\Phi_S \tag{B3}$$

$$\tan\alpha = \frac{2\cos\theta_S \sin 2\Phi_S}{(1+\cos^2\theta_S)\cos 2\Phi_S} \tag{B4}$$

$$\sin(2\psi+\alpha) = \frac{2\cos\alpha\tan\psi + \sin\alpha(1-\tan^2\psi)}{\tan^2\psi+1} \tag{B5}$$

$$\cos(2\psi+\alpha) = \frac{\cos\alpha(1-\tan^2\psi) - 2\sin\alpha\tan\psi}{\tan^2\psi+1} \tag{B6}$$

$$\tan(2\psi+\alpha) = \frac{2\tan\psi + \tan\alpha(1-\tan^2\psi)}{1-\tan^2\psi - 2\tan\alpha\tan\psi} \tag{B7}$$


[1] B. Abbott, R. Abbott, T. Abbott, M. Abernathy, F. Acernese, K. Ackley, C. Adams, T. Adams, P. Addesso, R. Adhikari, and et al., Physical Review Letters **116**, 10.1103/physrevlett.116.061102 (2016).

[2] T. L. S. Collaboration, J. Aasi, B. P. Abbott, R. Abbott, T. Abbott, and M. R. Abernathy, Classical and Quantum Gravity **32**, 074001 (2015).

[3] F. Acernese, M. Agathos, K. Agatsuma, D. Aisa, N. Allemandou, A. Allocca, J. Amarni, P. Astone, G. Balestri, G. Ballardin, F. Barone, J.-P. Baronick, M. Barsuglia, A. Basti, F. Basti, and T. S. B. et.al., Classical and Quantum Gravity **32**, 024001 (2014).

[4] The LIGO Scientific Collaboration, the Virgo Collaboration, the KAGRA Collaboration, R. Abbott, T. D. Abbott, F. Acernese, K. Ackley, *et al.*, arXiv e-prints , arXiv:2111.03606 (2021), arXiv:2111.03606 [gr-qc].

[5] K. Somiya, Classical and Quantum Gravity **29**, 124007 (2012).

[6] Y. Aso, Y. Michimura, K. Somiya, M. Ando, O. Miyakawa, T. Sekiguchi, D. Tatsumi, and H. Yamamoto (The KAGRA Collaboration), Phys. Rev. D **88**, 043007 (2013).

[7] T. Akutsu, M. Ando, K. Arai, Y. Arai, S. Araki, A. Araya, N. Aritomi, Aso, *et al.*, Progress of Theoretical and Experimental Physics **2021**, 05A101 (2021), arXiv:2005.05574 [physics.ins-det].

[8] M. Maggiore, C. V. D. Broeck, N. Bartolo, E. Belgacem, D. Bertacca, M. A. Bizouard, M. Branchesi, S. Clesse, S. Foffa, J. García-Bellido, S. Grimm, J. Harms, T. Hinderer, S. Matarrese, C. Palomba, M. Peloso, A. Ricciardone, and M. Sakellariadou, Journal of Cosmology and Astroparticle Physics **2020** (03), 050.

[9] M. Evans, R. X. Adhikari, C. Afle, S. W. Ballmer, S. Biscoveanu, S. Borhanian, D. A. Brown, Y. Chen, R. Eisenstein, A. Gruson, A. Gupta, E. D. Hall, R. Huxford, B. Kamai, R. Kashyap, J. S. Kissel, K. Kuns, P. Landry, A. Lenon, G. Lovelace, L. McCuller, K. K. Y. Ng, A. H. Nitz, J. Read, B. S. Sathyaprakash, D. H. Shoemaker, B. J. J. Slagmolen, J. R. Smith, V. Srivastava, L. Sun, S. Vitale, and R. Weiss, arXiv e-prints , arXiv:2109.09882 (2021), arXiv:2109.09882 [astro-ph.IM].

[10] S. Kawamura, M. Ando, N. Seto, S. Sato, M. Musha, I. Kawano, J. Yokoyama, T. Tanaka, K. Ioka, T. Akutsu, T. Takashima, K. Agatsuma, A. Araya, N. Aritomi, H. Asada, T. Chiba, S. Eguchi, M. Enoki, M.-K. Fujimoto, R. Fujita, T. Futamase, *et al.*, Progress of Theoretical and Experimental Physics **2021**, 05A105 (2021), arXiv:2006.13545 [gr-qc].

[11] P. Amaro-Seoane *et al.*, arXiv e-prints , arXiv:1702.00786 (2017), arXiv:1702.00786 [astro-ph.IM].

[12] C. Hanna, S. Caudill, C. Messick, A. Reza, S. Sachdev, L. Tsukada, K. Cannon, K. Blackburn, J. D. E. Creighton, H. Fong, P. Godwin, S. Kapadia, T. G. F. Li, R. Magee, D. Meacher, D. Mukherjee, A. Pace, S. Privitera, R. K. L. Lo, and L. Wade, Phys. Rev. D **101**, 022003 (2020).

[13] C. Messick, K. Blackburn, P. Brady, P. Brockill, K. Cannon, R. Cariou, S. Caudill, S. J. Chamberlin, J. D. E. Creighton, R. Everett, C. Hanna, D. Keppel, R. N. Lang, T. G. F. Li, D. Meacher, A. Nielsen, C. Pankow, S. Privitera, H. Qi, S. Sachdev, L. Sadeghian, L. Singer, E. G. Thomas, L. Wade, M. Wade, A. Weinstein, and K. Wiesner, Phys. Rev. D **95**, 042001 (2017).

[14] K. Cannon, S. Caudill, C. Chan, B. Cousins, J. D. E. Creighton, B. Ewing, H. Fong, P. Godwin, C. Hanna, S. Hooper, R. Huxford, R. Magee, D. Meacher, C. Messick, S. Morisaki, D. Mukherjee, H. Ohta, A. Pace, S. Privitera, I. de Ruiter, S. Sachdev, L. Singer, D. Singh, R. Tapia, L. Tsukada, D. Tsuna, T. Tsutsui, K. Ueno, A. Viets, L. Wade, and M. Wade, SoftwareX **14**, 100680 (2021), arXiv:2010.05082 [astro-ph.IM].

[15] T. Adams, D. Buskulic, V. Germain, G. M. Guidi, F. Marion, M. Montani, B. Mours, F. Piergiovanni, and G. Wang, Classical and Quantum Gravity **33**, 175012 (2016), arXiv:1512.02864 [gr-qc].

[16] F. Aubin, F. Brighenti, R. Chierici, D. Estevez, G. Greco,

G. M. Guidi, V. Juste, F. Marion, B. Mours, E. Nitoglia, O. Sauter, and V. Sordini, Classical and Quantum Gravity **38**, 095004 (2021), arXiv:2012.11512 [gr-qc].
[17] S. A. Usman, A. H. Nitz, I. W. Harry, C. M. Biwer, D. A. Brown, M. Cabero, C. D. Capano, T. D. Canton, T. Dent, S. Fairhurst, M. S. Kehl, D. Keppel, B. Krishnan, A. Lenon, A. Lundgren, A. B. Nielsen, L. P. Pekowsky, H. P. Pfeiffer, P. R. Saulson, M. West, and J. L. Willis, Classical and Quantum Gravity **33**, 215004 (2016).
[18] A. H. Nitz, T. Dent, T. D. Canton, S. Fairhurst, and D. A. Brown, The Astrophysical Journal **849**, 118 (2017).
[19] S. Klimenko and G. Mitselmakher, Classical and Quantum Gravity **21**, S1819 (2004).
[20] S. Klimenko, G. Vedovato, M. Drago, F. Salemi, V. Tiwari, G. A. Prodi, C. Lazzaro, K. Ackley, S. Tiwari, C. F. Da Silva, and G. Mitselmakher, Phys. Rev. D **93**, 042004 (2016).
[21] L. A. Wainstein and V. D. Zubakov, *Extraction of Signals from Noise* ((Prentice-Hall, Englewood Cliffs, NJ, 1970).
[22] B. J. Owen and B. S. Sathyaprakash, Phys. Rev. D **60**, 022002 (1999).
[23] The LIGO Scientific Collaboration, the Virgo Collaboration, the KAGRA Collaboration, R. Abbott, T. D. Abbott, F. Acernese, *et al.*, arXiv e-prints , arXiv:2111.03634 (2021), arXiv:2111.03634 [astro-ph.HE].
[24] T. A. Apostolatos, C. Cutler, G. J. Sussman, and K. S. Thorne, Phys. Rev. D **49**, 6274 (1994).
[25] X. Zhao, M. Kesden, and D. Gerosa, Phys. Rev. D **96**, 024007 (2017).
[26] C. Cutler and E. E. Flanagan, Phys. Rev. D **49**, 2658 (1994).
[27] K. S. Thorne, Rev. Mod. Phys. **52**, 299 (1980).
[28] L. E. Kidder, Phys. Rev. D **77**, 044016 (2008).
[29] G. Pratten, C. García-Quirós, M. Colleoni, A. Ramos-Buades, H. Estellés, M. Mateu-Lucena, R. Jaume, M. Haney, D. Keitel, J. E. Thompson, and S. Husa, Phys. Rev. D **103**, 104056 (2021), arXiv:2004.06503 [gr-qc].
[30] S. Ossokine, A. Buonanno, S. Marsat, R. Cotesta, S. Babak, T. Dietrich, R. Haas, I. Hinder, H. P. Pfeiffer, M. Pürrer, C. J. Woodford, M. Boyle, L. E. Kidder, M. A. Scheel, and B. Szilágyi, Phys. Rev. D **102**, 044055 (2020).
[31] V. Varma, S. E. Field, M. A. Scheel, J. Blackman, D. Gerosa, L. C. Stein, L. E. Kidder, and H. P. Pfeiffer, Phys. Rev. Res. **1**, 033015 (2019).
[32] C. Hoy, S. Fairhurst, M. Hannam, and V. Tiwari, Astrophys. J. **928**, 75 (2022), arXiv:2110.13542 [gr-qc].
[33] T. Damour, Phys. Rev. D **64**, 124013 (2001).
[34] E. Racine, Phys. Rev. D **78**, 044021 (2008).
[35] M. Hannam, P. Schmidt, A. Bohé, L. Haegel, S. Husa, F. Ohme, G. Pratten, and M. Pürrer, Phys. Rev. Lett. **113**, 151101 (2014).
[36] P. Schmidt, F. Ohme, and M. Hannam, Phys. Rev. D **91**, 024043 (2015).
[37] D. Gerosa, M. Mould, D. Gangardt, P. Schmidt, G. Pratten, and L. M. Thomas, Phys. Rev. D **103**, 064067 (2021), arXiv:2011.11948 [gr-qc].
[38] D. Gangardt, N. Steinle, M. Kesden, D. Gerosa, and E. Stoikos, Phys. Rev. D **103**, 124026 (2021).
[39] L. S. Finn and D. F. Chernoff, Phys. Rev. D **47**, 2198 (1993).
[40] R. Takahashi and T. Nakamura, The Astrophysical Journal **595**, 1039–1051 (2003).
[41] R. L. Forward, Phys. Rev. D **17**, 379 (1978).
[42] E. Poisson and C. M. Will, Phys. Rev. D **52**, 848 (1995).
[43] L. Bildsten and C. Cutler, Astrophys. J. **400**, 175 (1992).
[44] M. Kesden, D. Gerosa, R. O'Shaughnessy, E. Berti, and U. Sperhake, Phys. Rev. Lett. **114**, 081103 (2015).
[45] D. Gerosa, M. Kesden, U. Sperhake, E. Berti, and R. O'Shaughnessy, Phys. Rev. D **92**, 064016 (2015).
[46] S. Nissanke, D. E. Holz, S. A. Hughes, N. Dalal, and J. L. Sievers, Astrophys. J. **725**, 496 (2010), arXiv:0904.1017 [astro-ph.CO].
[47] L. S. Finn, Phys. Rev. D **46**, 5236 (1992).
[48] A. G. Wiseman, Phys. Rev. D **46**, 1517 (1992).
[49] A. Nitz, I. Harry, D. Brown, C. M. Biwer, J. Willis, T. D. Canton, C. Capano, T. Dent, L. Pekowsky, A. R. Williamson, S. De, M. Cabero, B. Machenschalk, D. Macleod, P. Kumar, S. Reyes, dfinstad, F. Pannarale, S. Kumar, T. Massinger, M. Tápai, L. Singer, G. S. C. Davies, S. Khan, S. Fairhurst, A. Nielsen, S. Singh, K. Chandra, shasvath, and veronica villa, gwastro/pycbc: v2.0.2 release of pycbc (2022).